

Solar O appendix

Sun · O · downloadable appendix

O.json v1.7 · 0 visible products · 7 quarantined entries

NOT_YET_DEFENSIBLE

No adopted Solar oxygen abundance is available from this feed.

Highlighted Products

No adopted highlighted product has been selected. No adopted Solar oxygen abundance is available from this feed.

Error-bar Forest Plot

No publication-eligible abundance products. Quarantined entries are documented below without plotted values.

How to read this forest. Each row is one independent product, preserving its band, holding, engine, indicator and line set. Engines and indicator families are never averaged. Solid bars show the supplied statistical uncertainty; wireframes show the supplied systematic uncertainty. The statistical basis and any incomplete error budget remain attached to the row. Grades describe the supplied line-pool provenance, not engine quality; evaluated theoretical CNO oscillator strengths are not relabelled as laboratory evidence. Quarantined, failed and unconstrained fits have no abundance mark. Experimental products remain explicitly not adopted. Literature references provide context, never tuning targets.

Solar oxygen context

Oxygen is traced by forbidden [O I], permitted O I and molecular OH indicators where available. Blends, departures from LTE and atmospheric structure affect these families differently, so their results retain separate identities.

[C/O indicator strategy](#) · [Methodology](#)

Open provenance decisions

O I 615.8nm: □ DISAGREES — reported to Ryan, not resolved. . Store log gf -0.409 (); summed pull -0.2966 (B+).

O I 630.0nm: □ DISAGREES — reported to Ryan, not resolved. □ store claims A, pull publishes B+. Store log gf -9.717 (A); summed pull -9.7744 (B+).

OPEN. Tachiev and NIST differ systematically by -0.016 dex on the 777 triplet and -0.005 dex on 8446. AGSS21 names Tachiev for N i only and states no O i gf source. Both values are reported in oi_tachiev_vs_nist.csv; NIST is what the store already carried and this script does not change it. Ryan decides.

[Adjudication record](#) · [Indicator gf checks](#)

Product visibility audit

Every feed entry is accounted for. Quarantined and archived entries retain their source disposition; their numeric fit outputs are available only in the labelled source download.

Withheld entries and source reasons (7)

quarantine · O · I · red-optical · iag_fts_solar_atlas · solar_iag · ALL · SET-AGSS21 · SYNTH · 1D-LTE RYA-1097 gated re-run precedence: this product does NOT pass the RYA-1092 eligibility gate, so it was published into quarantine and did not supersede anything. [NOT_YET_DEFENSIBLE] O I is outside the ratified real set {Fe I, Fe II, Al}. It is a pilot measurement, not a product.

quarantine · O · I · red-optical · iag_fts_solar_atlas · solar_iag · ALL · SYNTH · 1D-LTE

RYA-1097 gated re-run precedence: this product does NOT pass the RYA-1092 eligibility gate, so it was published into quarantine and did not supersede anything. [NOT_YET_DEFENSIBLE] O I is outside the ratified real set {Fe I, Fe II, Al}. It is a pilot measurement, not a product.

quarantine · O · I · red-optical · kpno_solar_atlas · solar_kpno_kurucz2005_corrected · ALL · SET-AGSS21 · SYNTH · 1D-LTE

RYA-1097 gated re-run precedence: this product does NOT pass the RYA-1092 eligibility gate, so it was published into quarantine and did not supersede anything. [NOT_YET_DEFENSIBLE] O I is outside the ratified real set {Fe I, Fe II, Al}. It is a pilot measurement, not a product.

quarantine · O · I · red-optical · kpno_solar_atlas · solar_kpno_kurucz2005_corrected · ALL · SYNTH · 1D-LTE
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quarantine · O · I · red-optical · kpno_solar_atlas · solar_kpno_molecfits_corrected · ALL · SET-AGSS21 · SYNTH · 1D-LTE

RYA-1097 gated re-run precedence: this product does NOT pass the RYA-1092 eligibility gate, so it was published into quarantine and did not supersede anything. [NOT_YET_DEFENSIBLE] O I is outside the ratified real set {Fe I, Fe II, Al}. It is a pilot measurement, not a product.

quarantine · O · I · red-optical · kpno_solar_atlas · solar_kpno_molecfits_corrected · ALL · SYNTH · 1D-LTE
RYA-1097 gated re-run precedence: this product does NOT pass the RYA-1092 eligibility gate, so it was published into quarantine and did not supersede anything. [NOT_YET_DEFENSIBLE] O I is outside the ratified real set {Fe I, Fe II, Al}. It is a pilot measurement, not a product.

quarantine · O · I · VIS · harps · solar_harps_molecfits_corrected · ALL · FORB-OI_6300 · SYNTH · 1D-LTE
RYA-1097 gated re-run precedence: this product does NOT pass the RYA-1092 eligibility gate, so it was published into quarantine and did not supersede anything. [STAT_BASIS_MISMATCH] sigma_stat here is a 'curvature_sigma' (SYNTH route, band_products.py:506) while the rest of the feed publishes a 'standard_error' (error_budget.py:609). The feed carries no stat_basis field, so `A +/- sigma\_stat` renders as comparable when it is not -- this product's bar reads ~sqrt(n) times worse than it is. [NOT_YET_DEFENSIBLE] O I is outside the ratified real set {Fe I, Fe II, Al}. It is a pilot measurement, not a product.

Download the evidence

[Download PDF — Solar O appendix](#)

[Source feed — includes quarantined fit outputs, not adopted abundances](#)

[Complete product visibility audit](#)

[Shared website/PDF report](#)

[Reproducibility manifest](#)

Reproducibility

Generated from the pinned CNO campaign snapshot. Source commit

a1ec12a9cc09472037e2af3b1242db2ce5b37c6b

Feed O.json v1.7 · updated 2026-09-13T04:21:45Z

Generated 2026-09-12T23:07:15-06:00 (source commit timestamp)

Feed SHA-256 dd90c8884d59e6ed90c3a637757e958476eb936e38fcb8fa7c5c4cf4702f1484

Generator scripts/generate_cno_publication.py v1.0.0

[Pinned scientific source](#)

References

Asplund, M.; Amarsi, A. M.; Grevesse, N. (2021). [The chemical make-up of the Sun: A 2020 vision](#). Solar atomic and molecular indicator context; no literature abundance is adopted here.